Optional Secondary flash or PSRAM (U4), using the second chip select (XIP_SS_N[1]) on GPIO0.
N.B. Pull-up R13 will be required as GPIOs default to pull-downs at power-up.

Make sure R7 and R8 are close to RP2350

=== Famicom bus interface (FC−EXPSYNTH additions) ===

J10: console−side gold fingers (1−30 front / 31−60 back)

J11: 60−pin slot (pin n −> pin n incl. PPU bus and /IRQ; /ROMSEL gated in synth mode)

U10−U12: 5V bus −> 3.3V (74LVC245, VCC=3V3, DIR=GND: B−>A, /OE=GND)

U13: D0−D7 bidirectional (74LVC245 at 3V3, 5V−tolerant B side; DIR=R/W: read A−>B, write B−>A for capture)

U14/U15: /ROMSEL gating and U13 output enable

RW_SYN = R/W AND NOT GAME (= NOT(RW_N OR GAME))
ROMSEL_EXP = ROMSEL_N OR RW_SYN (J11 pin44; game mode: = ROMSEL_N)
RDSEL_N = ROMSEL_N OR NOT R/W ; OE1_N = RDSEL_N OR EN_N
OE_N = NOT(RW_N OR OE1) (U17: U13 on for every CPU write, and for $8000+ reads when EN)

EN_N: GPIO27 low = U13 enabled, 10k pull-up keeps it disabled at reset.
GAME: high (default, 10k pull-up) = game mode: top cart gets every /ROMSEL cycle. GPIO30 low = synth mode (top cart sees writes only).

Power: console +5V and USB VBUS −> Schottky OR −> VSYS −> 3.3V LDO (IOVDD is up whenever console 5V is present)

D10: console +5V −> VSYS D11: VBUS −> VSYS (blocks back−feed)

+5V sense (GPIO40/ADC0)

Status LED, microSD (SPI0: GPIO32−35, CD=GPIO36), spare header

J14: 6 spare GPIO (GPIO31/43−47 left unrouted; GPIO30 = GAME mode select); GPIO0 is left free (PSRAM CS option via R11)

Takara (RP2350B core: Raspberry Pi Ltd minimal design REV2)
Sheet: /
File: FC−EXPSYNTH.kicad_sch

Title: FC−EXPSYNTH: Famicom expansion−audio relay cartridge (RP2350B)

Size: A1 Date: 2026−11−07 **Rev: E5**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 Id: 1/1